

YLIVIESKA, Finland

WMO: 027550

Lat: 64.050N

Lon: 24.725E

Elev: 76

StdP: 100.42

Time Zone: 2.00 (EUE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.2	-24.6	-31.2	0.2	-28.0	-27.2	0.3	-24.6	9.2	-3.6	8.2	-3.4	1.0	130	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	26.8	18.5	24.7	17.1	22.8	16.3	19.8	24.7	18.6	22.9	17.4	21.2	3.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.3	21.9	16.9	12.2	20.3	15.6	11.2	19.2	57.0	24.8	52.9	22.8	49.1	21.1	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.9	6.2	DB	-30.3	28.6	4.0	2.5	-33.2	30.4	-35.5	31.9	-37.8	33.3	-40.7	35.1	(4)
(5)			WB	-30.1	21.1	3.8	2.1	-32.8	22.6	-35.1	23.8	-37.2	25.0	-40.0	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-8.7	-8.5	-4.6	1.9	8.2	12.1	15.5	13.7	9.2	3.0	-0.9	-4.5	(6)	
	DBStd	9.69	7.46	7.70	5.80	3.76	4.38	3.65	3.46	3.47	3.54	4.12	4.87	6.30	(7)	
	HDD10.0	2964	581	518	454	242	88	19	2	8	57	219	328	448	(8)	
	HDD18.3	5588	839	752	712	492	316	191	102	149	276	476	578	707	(9)	
	CDD10.0	440	0	0	0	1	32	81	172	122	31	2	0	0	(10)	
	CDD18.3	21	0	0	0	0	1	3	13	4	0	0	0	0	(11)	
	CDH23.3	332	0	0	0	0	29	56	189	57	0	0	0	0	(12)	
	CDH26.7	62	0	0	0	0	2	12	42	6	0	0	0	0	(13)	
(14) Wind	WSAvg	2.9	2.7	2.9	3.0	3.0	3.2	3.1	2.7	2.6	2.8	2.9	3.0	3.1	(14)	
(15) Precipitation	PrecAvg	562	39	30	24	30	47	53	70	71	54	50	50	43	(15)	
	PrecMax	694	73	61	50	66	81	100	122	135	92	90	90	96	(16)	
	PrecMin	441	8	11	6	9	7	9	15	24	21	6	19	11	(17)	
	PrecStd	73	22	14	13	15	23	26	31	30	23	26	18	22	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.5	5.4	9.2	18.2	26.2	28.6	30.2	27.5	20.9	14.1	9.8	5.7	(19)	
		MCWB	2.6	3.3	4.9	10.3	15.0	18.6	19.7	20.1	15.4	11.6	8.7	5.2	(20)	
	2%	DB	2.2	3.6	6.4	14.2	23.6	24.7	27.7	25.0	18.9	11.6	8.1	4.4	(21)	
		MCWB	1.7	2.1	3.2	7.9	14.4	16.2	19.5	17.8	14.8	10.1	7.0	3.7	(22)	
	5%	DB	1.2	1.7	4.2	11.4	20.3	22.2	25.2	22.9	16.8	10.1	6.3	3.4	(23)	
		MCWB	0.7	0.9	1.8	6.3	12.8	15.1	18.1	16.8	13.3	8.9	5.8	2.7	(24)	
	10%	DB	0.3	0.5	2.7	9.1	17.1	20.1	23.4	20.9	15.2	8.8	4.9	2.1	(25)	
		MCWB	-0.1	0.0	0.9	4.9	10.7	14.0	17.3	15.9	12.6	7.7	4.4	1.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	3.6	5.4	11.0	16.8	19.5	22.0	21.1	16.7	12.1	8.8	5.3	(27)	
		MCDB	3.3	5.1	8.9	17.2	23.6	25.6	26.6	25.8	19.0	13.3	9.6	5.4	(28)	
	2%	WB	1.9	2.2	3.5	8.7	15.0	17.9	20.4	19.3	15.2	10.2	7.0	3.9	(29)	
		MCDB	2.1	3.4	5.7	13.5	21.7	23.2	25.4	23.2	18.1	11.2	7.8	4.2	(30)	
	5%	WB	0.8	1.1	2.3	6.6	13.4	16.1	19.2	17.7	14.0	9.0	5.7	2.7	(31)	
		MCDB	1.1	1.6	3.8	10.6	19.8	20.8	23.9	20.9	16.3	9.9	6.2	3.3	(32)	
	10%	WB	0.0	0.1	1.2	5.1	11.5	14.7	18.0	16.6	13.0	7.7	4.4	1.6	(33)	
		MCDB	0.4	0.5	2.3	8.5	16.1	18.9	22.5	19.5	14.9	8.5	4.8	2.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.7	9.7	10.2	11.9	11.5	11.7	11.1	9.3	6.2	4.4	5.4	(35)	
		MCDBR	4.2	5.3	9.3	15.1	18.0	16.1	15.2	14.8	12.3	7.0	4.9	4.5	(36)	
	5% WB	MCWBR	4.0	4.6	6.9	9.6	9.6	8.9	8.0	8.4	8.1	5.6	4.5	4.3	(37)	
		MCWBR	4.1	4.9	8.3	13.8	16.8	13.2	13.3	11.6	10.6	6.5	4.8	4.2	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.217	0.281	0.310	0.350	0.346	0.349	0.369	0.353	0.329	0.308	0.357	0.355	(40)	
		taud	1.987	2.140	2.199	2.308	2.443	2.460	2.422	2.460	2.514	2.375	2.620	2.623	(41)	
	Edn at Noon	Edn at Noon	350	622	770	824	867	870	837	819	767	605	324	173	(42)	
		Edh at Noon	27	61	89	101	96	97	98	87	67	50	25	15	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.14	0.69	2.17	3.74	4.89	5.40	5.02	3.79	2.15	0.80	0.19	0.06	(44)	
	RadStd	RadStd	0.01	0.13	0.21	0.35	0.52	0.47	0.49	0.38	0.23	0.14	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (45 neighbors)		+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page